

Experiment 3

Name:Pratik Chavan

Div/Batch:A/A1 Roll No.07

.MODEL SMALL

.STACK 100H

.DATA

 ARRAY DB 10, 25, 15, 40, 5, 30, 50, 20

 LEN EQU \$ - ARRAY

 MIN DB 0

 MAX DB 0

.CODE

MAIN PROC

 MOV AX, @DATA

 MOV DS, AX

 MOV SI, 0

 MOV AL, ARRAY[SI]

 MOV MIN, AL

 MOV MAX, AL

FIND_MIN_MAX:

 MOV AL, ARRAY[SI]

 CMP AL, MAX

 JG UPDATE_MAX

 CMP AL, MIN

JL UPDATE_MIN

JMP NEXT_ELEMENT

UPDATE_MAX:

MOV MAX, AL

JMP NEXT_ELEMENT

UPDATE_MIN:

MOV MIN, AL

NEXT_ELEMENT:

INC SI

CMP SI, LEN

JL FIND_MIN_MAX

MOV AX, 4C00H

INT 21H

MAIN ENDP

END MAIN

OUTPUT:

The screenshot shows a debugger window with the following content:

Address	Disassembly	Comment	Register/Value
cs:000D	8A840000	+ mov al, [array + si]	ax 0C07
cs:0011	3AC4	+ cmp al, ah ; Compare c	bx 0000
cs:0013	7F15	+ jg update_max ; If AL	cx 0000
cs:0015	3AC0	+ cmp al, al ; Compare c	dx 0000
cs:0017	7C0D	+ jl update_min ; If AL	si 0006
cs:0019	46	+ inc si	di 0000
cs:001A	A00700	+ mov al, [n] ; Load arr	bp 0000
cs:001D	3BF0	+ cmp si, ax ; Compare i	sp 0100
cs:001F	7CEC	+ jl find_min_max ; If s	ds 0B7F
cs:0021	B8004C	+ mov ax, 4C00h	es 0B6C
cs:0024	CD21	+ int 21h	ss 0B80
#max#update_min			cs 0B7C
			ip 001F

Registers (right side):

c=1
z=0
s=1
o=0
p=1
a=1
i=1
d=0

Stack (bottom):

es:0000	CD 20 7D 9D 00 EA FF FF = 1¥ 2
es:0008	AD DE 32 0B C5 05 6B 07 1 26 2k
es:0010	15 03 28 0B 15 03 93 01 3 03 00
es:0018	01 01 01 00 02 04 FF FF 88 8

Stack pointers (bottom right):

ss:0102	0403
ss:0100	52FB

extra

S. E. Sem IV / R-19 / FH 23 / 15/05/2023

(Comp) Duration: 3hrs

[Max Marks:80]

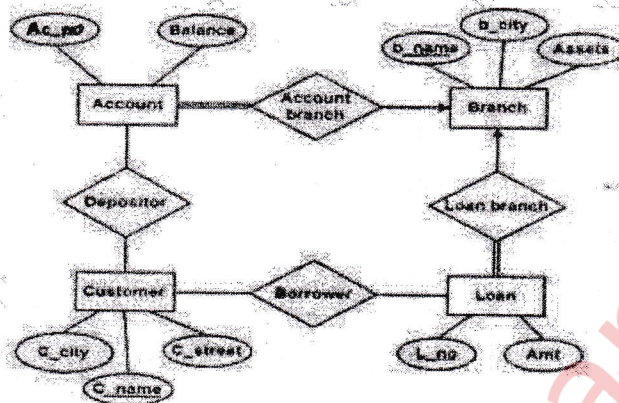
Q.P : 29883

- N.B. : (1) Question No 1 is Compulsory.
 (2) Attempt any three questions out of the remaining five.
 (3) All questions carry equal marks.
 (4) Assume suitable data, if required and state it clearly.

1 Attempt any **FOUR**

[20]

- a Identify different users of database management system
 b Convert following E-R diagram to relational schema



- c Explain all types of integrity constraints with an examples?
 d List all functional dependencies satisfied by the relation.

X	Y	Z
X1	Y1	Z1
X1	Y2	Z1
X2	Y2	Z1
X2	Y2	Z1

- e Discuss Log based recovery with an example
- 2 a Discuss three layer schema architecture with suitable diagram. What is Data Independence? Explain types of data independence. [10]
 b What is deadlock? Give deadlock prevention methods with suitable example [10]
- 3 a Construct an ER diagram and convert it into a relational model for a company which has several employees working on different types of Projects. Several employees are working for one department, every department has a manager. Several employees are supervised by one employee. Employees have zero or more dependents [10]

29883

- b Explain the following Relational Algebra operations with suitable example. [10]
- 1) Generalized Project
 - 2) Select
 - 3) Union
 - 4) Rename
 - 5) Natural Join
- 4 a Write SQL queries for the given database [10]
- Book(book_id, title, author, cost)
- Store(store_no, city, state, inventory_val)
- Stock(store_no, book_id, quantity)
- (i) Modify the cost of DBMS books by 10%
 - (ii) Find the total number of books in Mumbai stores
 - (iii) Find title of all books whose title contains the word 'System'
 - (iv) Find title of the most expensive book
 - (v) Add a new record in Book (Assume values as per requirement)
- b Why there is need of normalization? Explain 1NF, 2NF, 3NF and BCNF with example. [10]
- 5 a Describe ACID properties with examples [10]
- b Give example of serial schedule and equivalent to serial schedule with respect to conflict serializability. Discuss conflict serializability with example [10]
- 6 Write short note on the following (Any four) [20]
- a Conversion of Specialization to relational schema with suitable example [05]
 - b Types of attributes [05]
 - c 2PL concurrency control protocol [05]
 - d Triggers [05]
 - e Lossless decomposition [05]

[Time: 3 Hours]

[Marks: 80]

- N.B. : (1) Question No 1 is Compulsory.
 (2) Attempt any three questions out of the remaining five.
 (3) All questions carry equal marks.
 (4) Assume suitable data, if required and state it clearly.

1 Attempt any FOUR

[20]

- a Compare File Processing System with Database Management system

05

b

05

T1	T2
read(A) A := A - 50	
	read(A) temp := A * 0.1 A := A - temp write(A) read(B)
write(A) read(B) B := B + 50 write(B) commit	
	B := B + temp write(B) commit

Draw the precedence graph for above schedule?

- c Define with an example different type of Entities in ER diagram
- d Define Triggers. Write syntax and example of trigger.
- e Explain five aggregate functions of SQL with example?

05

05

05

- 2 a Design an EER diagram for Hospital Management System. And map it into relational model. Assume Suitable data.

[10]

- b Brief overall database architecture with suitable diagram.

[10]

- 3 a Consider the following employee database.

[10]

Employee (empname, street, city, date_of_joining)

Works (empname, company_name, salary)

Company (company_name, city)

Manages (empname, manager_name)

Write the SQL queries for each of the statements given below

- a) Modify the database so that 'John' now lives in 'Mumbai'.
b) Find all employees who joined in the month of October.
c) Give all employees of 'ABC Corporation' a 10% raise.
d) Find all employees in the database who live in the same cities as the companies for which they work
e) Find all employees who earn more than average salary of all employees of their company
- b Explain following relational algebra operators with example [10]
a) Selection operator b) Union operator
c) Rename operator d) Cartesian product
- 4 a Explain concurrency control and explain time Stamp based protocol of concurrency control. [10]
b Why there is need of normalization? Explain 1NF,2NF,3NF and BCNF with examples. [10]
- 5 a Describe ACID properties with examples and explain state transition diagram of transaction. [10]
b What is Deadlock. Explain wait-die and wound-wait methods with suitable example. [10]
- 6 **Attempt any two**
a Explain in detail with example of conflict and view serializability . [10]
b Explain following Integrity constraints: [10]
a) Key Constraints.
b) Domain Constraints (Null & Default Constraints).
c) Referential Constraints.
d) Check Constraints.
c Write short note on Log based recovery mechanism [10]
-

Q4. Drawback of traditional file processing system. Or explain disadvantages of conventional file-based system compared to database management system.

Q5. Explain advantages and disadvantages of DBMS.

Q6. List out the difference between file processing and DBMS.

Q7. Write in detail about applications of DBMS.

Q10. Explain data independence.

Q11. Write about DBMS system architecture and components of DBMS.

Q12. Write about Database Administrator and functions of database administrations.

Q2. Explain E-R model and its components. Explain weak and strong entity set.

Q3. Explain relationships and its degree.

Q4. List symbols used in ER diagram and its representations.

Q5. Explain mapping cardinality in ER diagram and participation constraints & its types.

Q7. Define attributes and its types. Explain relationship attributes.

Q9. Explain all keys used in DBMS.

Q11. Explain generalization, specialization, aggregation and constraints on generalization.

Q4. Explain tuple, table, attribute, domain, and properties of relational database.

Q5. Explain CODD's rule (All 12 rules) in detail.

Q6. Explain relational schema.

Q7. Explain types of keys in DBMS with suitable example.

Q9. Explain selection and projection operators in relational algebra with suitable example.

Q10. Explain all types of join operator with suitable example in relational algebra.

Sample 1 (General Text):

"The quick brown fox jumps over the lazy dog. This sentence contains every letter of the English alphabet. Testing text extraction and conversion features requires diverse content, including punctuation, numbers (1234567890), and special characters (!@#\$%^&*()). Let's see how well this works!"

Sample 2 (Longer Paragraph):

"Artificial intelligence is rapidly transforming various industries, from healthcare to finance. Machine learning models analyze vast amounts of data to generate insights, automate tasks, and improve efficiency. Speech-to-text technology enables seamless communication, while text-to-speech tools enhance accessibility. Ensuring accurate extraction of text from documents remains a crucial challenge in modern computing."

Sample 3 (Dialogue):

Alice: "Hey Bob, have you tried the new text extraction tool?"

Bob: "Yes! It works surprisingly well. I extracted text from a complex PDF in seconds!"

Alice: "That's awesome! Can it convert text to speech too?"

Bob: "Absolutely. It even supports multiple languages!"

Sample 4 (Multilingual Text for Translation Tests):

- English: "Hello, how are you?"
- Spanish: "Hola, ¿cómo estás?"
- French: "Bonjour, comment ça va?"
- German: "Hallo, wie geht es dir?"

Sample 5 (Numbers, Dates, and Special Characters):

"Invoice No: 2025-0319

Total Amount: \$1,250.75

Due Date: 25th March 2025

Contact: support@example.com

Reference ID: #ABCD1234XYZ"

Artificial intelligence is rapidly transforming various industries, from healthcare to finance. Machine learning models analyse vast amounts of data to generate insights, automate tasks, and improve efficiency. Speech-to-text technology enables seamless communication on, while text-to-speech tools enhance accessibility. Ensuring accurate extract on of text from documents remains a crucial challenge in modern computing.

Hello this is a test for our new project pdf utilizer